

Feature Engineering & Model Building

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After completing preprocessing and exploratory analysis, feature engineering and model experimentation were performed to identify the optimal feature subset and regression model for residential house price prediction.

Feature Encoding

Categorical variables were converted into numerical format using One-Hot Encoding (`pd.get_dummies`). Encoding enabled machine learning algorithms to process categorical house attributes effectively.

Feature Selection Techniques

Multiple feature selection approaches were evaluated to determine the best feature subset.

1. SelectKBest

SelectKBest was applied to select top-ranked features based on statistical relationship with the target variable.

Different K values were tested to observe model performance variation.

2. Principal Component Analysis (PCA)

PCA was applied to reduce dimensionality while preserving maximum variance.

Several component combinations were evaluated.

Example:

PCA_5

PCA_6

PCA_8

PCA transformed original variables into lower-dimensional components for model training.

3. Recursive Feature Elimination (RFE)

RFE recursively removed less important variables and retained only influential features.

Different feature counts were tested to identify the optimal subset.

RFE later produced the highest performing feature combination.

Model Development

For each feature selection approach, multiple regression algorithms were trained and compared.

Models evaluated:

- Linear Regression
- Support Vector Machine (Linear Kernel)
- Support Vector Machine (Non-Linear Kernel)
- Decision Tree Regressor
- Random Forest Regressor

To simplify experimentation and maintain reusable code, model functions were organized inside:

Regression_utils.py

This utility file centralized training and evaluation functions for future projects.

Hyperparameter Optimization

After identifying promising models, RandomizedSearchCV was applied for hyperparameter tuning.

Randomized Search was selected to reduce computational time while exploring multiple parameter combinations.

Example optimized parameters:

n_estimators

max_depth

Model Comparison

Performance of all feature selection methods and algorithms was compared.

Methods evaluated:

SelectKBest

PCA

RFE

Algorithms evaluated:

Linear

SVM Linear

SVM Non-Linear

Decision Tree

Random Forest

The comparison showed that:

RFE + Random Forest

generated the strongest predictive performance.

Final Selected Configuration

Feature Selection : RFE

Algorithm : Random Forest Regressor

Hyperparameter : RandomizedSearchCV

This selected configuration was forwarded to Folder 5 for final model creation and deployment.
